Exp no : 1(a)

Aim : Analyze the trend of data science job postings over the last decade.

Description : Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code :

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}


df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

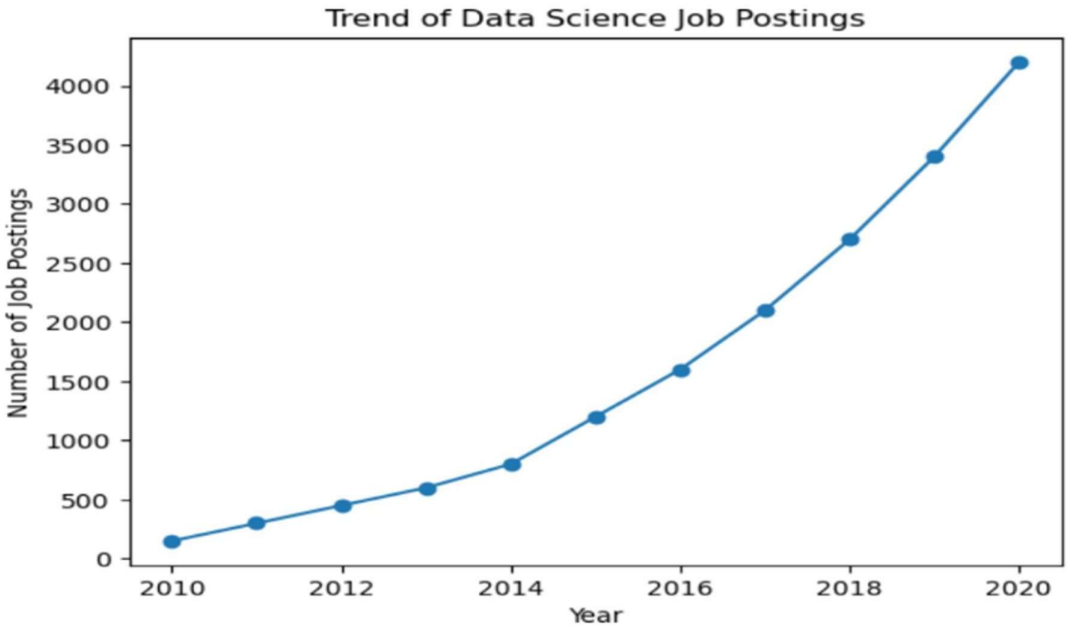
plt.ylabel('Number ofJob

Postings') plt.show()
```

Sample data input :

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200



RESULT: The trend of data science job posting over the last decade is analysed.

EXP NO: 1(b)

AIM: Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

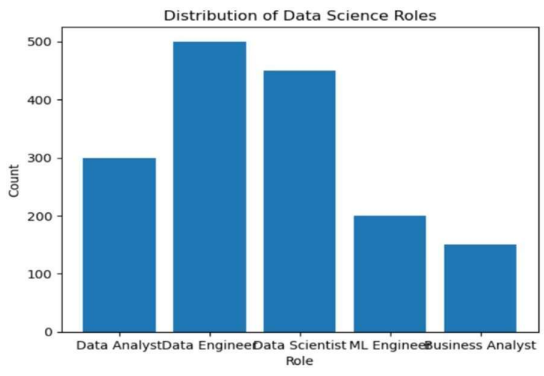
DESCRIPTION: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

SAMPLE INPUT:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst. counts = 300, 500, 450, 200, 150.

SAMPLE OUTPUT:



RESULT: The distribution of various datascience roles from a dataset is visualized.

EXP NO: 1(c)

AIM: Conduct an experiment to differentiate Structured , Un-structured and Semi-structured data based on data sets given.

DESCRIPTION: Create small datasets for each type and explain their characteristics.

PROGRAM:

```
# Structured data example

structured_data =

pd.DataFrame({ 'ID': [1, 2, 3],

'Name': ['Alice', 'Bob', 'Charlie'],

'Age': [25, 30, 35]

})

print("Structured Data:\n", structured_data)


# Unstructured data example

unstructured_data = "This is an example of unstructured data. It can be a piece
of text, an image,

or a video file."

print("\nUnstructured Data:\n", unstructured_data)


# Semi-structured data example (JSON)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165,

'Weight': 68}} print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

OUTPUT:

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}
}
```

RESULT: The experiment to differentiate structured, un-structured and semi-structured data based on datasets given are conducted.

EXP NO: 1(d)

AIM: Conduct an experiment to encrypt and decrypt given sensitive data.

DESCRIPTION: Use the cryptography library to encrypt and decrypt a piece of data.

PROGRAM:

```
# Generate key and encrypt data
from cryptography.fernet import
Fernet key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering
College' key =
Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:",
decrypted_text)
```

OUTPUT:

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

RESULT: The experiment to encrypt and decrypt given sensitive data is conducted.